



# THE BOXES

Research dossier

A historical drama set in 1945 when the mathematician John von Neumann invented the modern computing blueprint.

In 1945, theoretical mathematicians and military engineers raced to build electronic calculating machines for wartime weapons programs. Research records trace the daily movement between secure university laboratories in New Jersey and desert checkpoints in New Mexico. The story centers on a mid-level military stenographer assigned to record these scientific sessions. This bystander watches John von Neumann approach worldwide survival as an abstract mathematical game.

| Depth               | Scout                      |
|---------------------|----------------------------|
| Research confidence | 90% <div><div></div></div> |
| Coverage            | 94% <div><div></div></div> |
| Status              | —                          |
| Boxes               | 6                          |
| Evidence fragments  | 69                         |
| Distinct sources    | 55                         |
| Distinct domains    | 35                         |
| Contradictions      | 0 conflict, 0 context      |
| Research passes     | 2                          |
| Generated           | 2026-09-09 14:27 UTC       |
| Project id          | demo                       |

## THE BOXES

*What the research found, in plain language, box by box. Every claim below is traceable to the sources listed under it.*

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### WARTIME TRANSIT AND SECURITY

97%  10 items · 7 domains

Manhattan Project records outline strict travel protocols connecting Princeton to the Los Alamos laboratory in 1945. Armed couriers moved physical bomb components and technical records under guard. Military checkpoints and barbed-wire perimeters separated research areas from surrounding towns. Historical travel logs track civilian commuters, train schedules, and rationing directives across the transit network.

- Security and the Manhattan Project
- Establishing Los Alamos, 1942-1943 - Manhattan Project
- MARC view: Manhattan District history Project Y the Los Alamos Project : vol. II, August 1945 through December 1946
- Then and Now: Secure Transportation | Department of Energy · 2026-05-15
- Los Alamos From Below: Reminiscences 1943-1945, by Richard Feynman
- First Commuters
- Memorandum from J. R. Oppenheimer to Brigadier General Farrell, May 11, 1945 | National Security Archive
- Manhattan Project: Atomic Rivals and the ALSOS Mission, 1938-1945

### INSTITUTIONAL WORKSPACES AND HARDWARE

97%  8 items · 5 domains

Archival records document the hardware installations at the Moore School of Electrical Engineering and the Institute for Advanced Study. Engineers J. Presper Eckert and John W. Mauchly constructed the ENIAC using thousands of electronic switches to compute numerical solutions. Technical files track the subsequent design of the EDVAC as a stored-program architecture. Director office records preserve blueprints, analyzer photographs, and equipment manifests from these facilities.

- Guide, Moore School of Engineering. Office of the Director Records (UPD 8.4) · 2019-11-18
- About the Archives
- Moore School Office of the Director Records · 2020-04-10
- ENIAC | Penn Engineering - University of Pennsylvania
- EDVAC - Wikipedia
- ENIAC Trial Exhibits Master Collection (Index)
- ENIAC - Wikipedia
- University of Pennsylvania 1946

### ELITE ACADEMIC SOCIAL LIFE

96%  17 items · 10 domains

Directories and photograph collections preserve the domestic life of Institute for Advanced Study faculty during the war years. University clubs maintained formal dinner menus, dress expectations, and cocktail service for resident scholars. Domestic staff managed academic households and served private gatherings. Archival images show the social gatherings that operated alongside defense research contracts.

- Institute for Advanced Study Faculty and Staff Directory 1943-1944
- Institute for Advanced Study (Princeton, N.J.) photographs: Other Individuals
- Cap and Gown Club-Constitutions and Menu, 1905-1946 - Finding Aids
- 'The servants of the Nation': Quiet Revolutions in Education
- amias\_brochure\_web\_2
- Image Collections - Family History Sources in the Princeton University Archives - Research Guides at Princeton University
- Princeton University Archives
- University Archives | Princeton University Library
- Princeton University Campus Dining - Today's Menus
- Menu - Winberie's Princeton
- Institute for Advanced Study - Wikipedia
- Key titles on the history of the Institute for Advanced Study
- Key and Seal Club - Miscellaneous, 1904-1967 - Finding Aids

## MECHANICAL COMPUTING ACOUSTICS

96%  11 items · 6 domains

Technical manuals describe the sound profiles of mid-century computing machinery. Mechanical punch-card sorters, accounting tabulators, and collators generated rhythmic metallic strikes during data sorting. Relays clicked continuously under heavy computational loads. Heavy cooling fans operated inside the machine rooms to manage equipment temperatures.

- [ibm :: punchedCard :: Training :: A24-1010-0 IBM Operators ... · 2013-01-09](#)
- [The IBM 407 Accounting Machine](#)
- [Reference manual IBM 82, 83, and 84 sorters; 1962](#)
- [Restoring the IBM 1401 - CHM - Computer History Museum · 2019-08-12](#)
- [IBM Card Sorters - Columbia University](#)
- [IBM Operators Guide : International Business Machines ...](#)
- [The punched card tabulator - IBM](#)
- [Punched card sorter - Wikipedia · 2026-03-25](#)

## MILITARY CLASSIFICATION AND INFORMATION CONTROLS · emergent

95%  9 items · 5 domains

Declassified histories outline the document security rules governing atomic and computational research in 1945. Military directives mandated strict document classification markings, sign-in logs, and burn-bag disposal procedures. Security offices investigated leaks and tracked unauthorized disclosures of technical data. Theoretical ballistics reports required compartmented access permissions and military counterintelligence oversight.

- [Manhattan District History Project Y The Los Alamos Project Vol. I Inception Until August 1945 \(Technical Report\) | OSTI.GOV](#)
- [Security and the Manhattan Project](#)
- [DOE OpenNet document, Drawing Back the Curtain of Secrecy, Restricted Data Declassification Decisions 1946 to the Present \(RDD-5\)](#)
- [History of Classification and Declassification \(DOE\)](#)
- [Department of Energy \(DOE\) OpenNet documents](#)
- [Security Classification of Information, volume 1 \(Quist\), Chapter Four](#)
- [Appendix L: Additional Classification and Secrecy Information](#)
- [WWII Atomic Bomb Project Had More Than 1,500 "Leaks" - Federation of American Scientists · 2024-06-26](#)
- [DOE OpenNet document, Drawing Back the Curtain of Secrecy, Restricted Data Declassification Decisions 1946 to the Present](#)

## MATHEMATICAL COLLABORATION DIALECT

84%  14 items · 8 domains

Technical reports from the Army Ballistic Research Laboratory record the specialized vocabulary of wartime ballisticians. Declassified correspondence shows mathematicians discussing weapon trajectories, differential equations, and shell dispersions. Meeting transcripts capture formal exchanges between military officers and academic consultants. The recorded phrasing reflects specific analytical targets set for wartime ordnance tests.

- [\[PDF\] Proving Grounds, 1940-67 - National Archives](#)
- [Ballisticians in War and Peace Volume I - DTIC](#)
- [From Science to Overmatch: A Case Study of the Ballistic Research Laboratory Prior to and during World War 2](#)
- [Ballisticians in War and Peace. A History of the United States Army Ballistics Research Laboratory. Volume 3. 1977-1992. | National Technical Reports Library - NTIS](#)
- [Ballisticians in War and Peace. A History of the United States Army Ballistics Research Laboratory. Volume 3. 1977-1992,](#)
- [AAF. See U.S. Army Air Forces Aberdeen Proving Ground ...](#)
- [U.S. Army Ballistic Research Laboratory](#)
- [Handbook of Ballistic and Engineering Data for Ammunition ...](#)
- [Papers of John Von Neumann \[finding aid\]. Library of Congress ...](#)
- [Von Neumann in the United States | Springer Nature Link](#)
- [John Von Neumann and Klaus Fuchs papers, | Library of Congress](#)

## CONTRADICTIONS

*Where two sources disagree. Similarity alone never decides this; each pair below was read and judged.*

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None found.

## REFERENCE REEL

*The strongest material, cut into a sequence a director can read.*

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### 00:00 Wartime Secrecy Outpost

Los Alamos operated as a heavily fortified wartime research compound during the Manhattan Project.

- Security and the Manhattan Project · [irp.fas.org](http://irp.fas.org)
- Establishing Los Alamos, 1942-1943 - Manhattan Project · [osti.gov](http://osti.gov)
- MARC view: Manhattan District history Project Y the Los Alamos Project : vol. II, August 1945 through December 1946 · [catalog.hathitrust.org](http://catalog.hathitrust.org)
- Then and Now: Secure Transportation | Department of Energy · [stage.energy.gov](http://stage.energy.gov) · 2026-05-15
- Los Alamos From Below: Reminiscences 1943-1945, by Richard Feynman · [calteches.library.caltech.edu](http://calteches.library.caltech.edu)
- First Commuters · [lanl.gov](http://lanl.gov)

### 00:15 Princeton Scholarly Life

The Institute for Advanced Study and Princeton University provided a quiet academic setting for theoretical research.

- About the Archives · [ias.edu](http://ias.edu)
- Institute for Advanced Study Faculty and Staff Directory 1943-1944 · [albert.ias.edu](http://albert.ias.edu)
- Institute for Advanced Study (Princeton, N.J.) photographs: Other Individuals · [albert2.ias.edu](http://albert2.ias.edu)
- Cap and Gown Club-Constitutions and Menu, 1905-1946 - Finding Aids · [findingaids.princeton.edu](http://findingaids.princeton.edu)
- 'The servants of the Nation': Quiet Revolutions in Education · [jstor.org](http://jstor.org)
- [amias\\_brochure\\_web\\_2](#) · [ias.edu](http://ias.edu)

### 00:30 Moore School Computing

Engineers at the University of Pennsylvania constructed early electronic calculating machines including the ENIAC.

- Guide, Moore School of Engineering. Office of the Director Records (UPD 8.4) · [archives.upenn.edu](http://archives.upenn.edu) · 2019-11-18
- Moore School Office of the Director Records · [archives.upenn.edu](http://archives.upenn.edu) · 2020-04-10
- ENIAC | Penn Engineering - University of Pennsylvania · [engineering.upenn.edu](http://engineering.upenn.edu)
- ENIAC Trial Exhibits Master Collection (Index) · [archives.upenn.edu](http://archives.upenn.edu)
- ENIAC - Wikipedia · [en.wikipedia.org](http://en.wikipedia.org)
- University of Pennsylvania 1946 · [r2.ieee.org](http://r2.ieee.org)

### 00:45 Stored Program Architecture

The conceptual design for the EDVAC established the blueprint for modern electronic computers.

- EDVAC - Wikipedia · [en.wikipedia.org](http://en.wikipedia.org)
- The IBM 407 Accounting Machine · [columbia.edu](http://columbia.edu)

## DEPARTMENTS

*The same boxes, grouped by which crew briefs off them, with what each one actually contains.*

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### Script

WARTIME TRANSIT AND SECURITY · 10 items

Manhattan Project records outline strict travel protocols connecting Princeton to the Los Alamos laboratory in 1945. Armed couriers moved physical bomb components and technical records under guard. Military checkpoints and barbed-wire perimeters separated research areas from surrounding towns. Historical travel logs track civilian commuters, train schedules, and rationing directives across the transit network.

MATHEMATICAL COLLABORATION DIALECT · 14 items

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MILITARY CLASSIFICATION AND INFORMATION CONTROLS · 9 items

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### Casting

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### Costume

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The GRUBHUB logo is displayed within a red rectangular box. The word "GRUBHUB" is written in a bold, white, sans-serif typeface.

### Art direction

INSTITUTIONAL WORKSPACES AND HARDWARE · 8 items

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## Sound

MECHANICAL COMPUTING ACOUSTICS · 11 items

Technical manuals describe the sound profiles of mid-century computing machinery. Mechanical punch-card sorters, accounting tabulators, and collators generated rhythmic metallic strikes during data sorting. Relays clicked continuously under heavy computational loads. Heavy cooling fans operated inside the machine rooms to manage equipment temperatures.

## Locations

INSTITUTIONAL WORKSPACES AND HARDWARE · 8 items

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## PRIOR ART

44 existing films surveyed for a similar premise, seeded from: mathematician, invention, cold war, computer science, atomic bomb.

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### To End All War: Oppenheimer & the Atomic Bomb (2023)

similarity 0.68 · <https://www.themoviedb.org/movie/1149947>

### The Imitation Game (2014)

similarity 0.67 · <https://www.themoviedb.org/movie/205596>

### Oppenheimer (2023)

similarity 0.66 · <https://www.themoviedb.org/movie/872585>

### A Beautiful Mind (2001)

similarity 0.64 · <https://www.themoviedb.org/movie/453>

### The Current War (2018)

similarity 0.62 · <https://www.themoviedb.org/movie/418879>

### 23 (1998)

similarity 0.56 · <https://www.themoviedb.org/movie/1557>

### Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1964)

similarity 0.55 · <https://www.themoviedb.org/movie/935>

### The Man from U.N.C.L.E. (2015)

similarity 0.54 · <https://www.themoviedb.org/movie/203801>

### Mayday (2026)

similarity 0.54 · <https://www.themoviedb.org/movie/1137844>

### Bridge of Spies (2015)

similarity 0.53 · <https://www.themoviedb.org/movie/296098>

### The Shape of Water (2017)

similarity 0.52 · <https://www.themoviedb.org/movie/399055>

### The Lives of Others (2006)

similarity 0.52 · <https://www.themoviedb.org/movie/582>

### Where this stands, and where it does not yet

Originality is claimed only relative to the titles above, never absolutely.

A mid-level military stenographer acts as a bystander observing von Neumann treat global survival as an abstract logic game.

None of the comparison films combine a bystander perspective with a satirical tone and an open bureaucratic stalemate ending.  
checked against: Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1964), Oppenheimer (2023), The Shape of Water (2017), The Lives of Others (2006)

→ Center the 1945 computing conferences on the civilian clerks who transcribe the architecture while realizing the scientists view humanity as raw numbers.



**An external investigator audits von Neumann for mathematical fraud and discovers the computing blueprint hides deliberate hardware vulnerabilities.**

None of the comparison films feature an investigator finding an engine of exploitation built into hardware design ending in institutional coverup.

checked against: The Current War (2018), The Imitation Game (2014), Bridge of Spies (2015), 23 (1998)

→ Show an army engineer discovering that von Neumann designed the stored-program architecture to grant permanent back-channel access to weapons laboratories.

**A human computer calculator feels victimized when her manual calculation role is automated away by the architecture.**

None of the comparison films frame the dawn of computation around the displaced labor victim with a tragic tone ending in complete displacement.

checked against: The Imitation Game (2014), A Beautiful Mind (2001), To End All War: Oppenheimer & the Atomic Bomb (2023)

→ Follow a Los Alamos human computer as von Neumann uses her daily mathematical labor to program the machine that abolishes her profession.

## RESEARCH LOG

*What each pass searched, found, and did next. Process detail, not required reading.*

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### PASS 000

40 sources · 46 fragments · 10 via Extract

coverage 0% → 90% confidence 0% → 87%

INSTITUTIONAL WORKSPACES AND HARDWARE — Moore School University of Pennsylvania ENIAC EDVAC laboratory photographs 1945 equipment blueprints

WARTIME TRANSIT AND SECURITY — Manhattan Project courier travel regulations train Princeton Los Alamos wartime security 1945

ELITE ACADEMIC SOCIAL LIFE — Institute for Advanced Study Princeton faculty social life dinner parties domestic servants 1940s diaries

MECHANICAL COMPUTING ACOUSTICS — IBM punched card sorter accounting machine relay operating acoustic sound recordings 1940s

MATHEMATICAL COLLABORATION DIALECT — Aberdeen Proving Ground Ballistic Research Laboratory computing meeting minutes correspondence 1944 1945

→ investigate MATHEMATICAL COLLABORATION DIALECT (score 0.74)

### PASS 001

24 sources · 24 fragments · 6 via Extract

coverage 90% → 94% confidence 87% → 90%

opened: MILITARY CLASSIFICATION AND INFORMATION CONTROLS

ELITE ACADEMIC SOCIAL LIFE — Princeton faculty social life diaries cocktails dinner menus 1940s Institute for Advanced Study

MATHEMATICAL COLLABORATION DIALECT — Ballistic Research Laboratory Aberdeen Proving Ground correspondence computing jargon 1944 1945 von Neumann

MILITARY CLASSIFICATION AND INFORMATION CONTROLS — Manhattan Project Manhattan District History declassified security regulations document classification stamping burn bag procedures 1945

→ investigate MATHEMATICAL COLLABORATION DIALECT (score 0.84)